

Displacive phase transitions

12

12.1 Introduction to displacive phase transitions

Many applications of the solid state sciences exploit the electronic or mechanical properties of solids. The important materials are metals or semiconducting materials. These materials tend to have relatively simple crystal structures, such as the cubic and hexagonal structures we met in Chapter 2 (fcc, hcp, diamond, NaCl, and ZnS structures), and interest is focused more on the behaviour of the electrons than on the behaviour of the atoms *per se*. On the other hand, there are many properties of materials that depend on some degree of complexity of the crystal structure, and on its anisotropy (Chapter 7), and these can be significantly enhanced if the material undergoes a phase transition at some temperature. This is particularly true for dielectric, pyroelectric, and piezoelectric properties. Moreover, **displacive phase transitions** are associated with important phenomena such as high-temperature superconductivity in cuprate ceramics.

Not only are phase transitions of considerable technological importance, it is also true that the study of phase transitions is particularly enjoyable because it brings together many of the topics we have discussed as separate topics in earlier chapters. In particular, the study of phase transitions brings together issues of crystal structures, symmetry, bonding, thermodynamics, and lattice dynamics, including the need to include anharmonic interactions. Because of this, it is appropriate to close this book with a survey of phase displacive transitions in crystalline materials.

We met a number of phase transitions in Chapter 2. It can be argued that the most important examples are the perovskites, which have displacive phase transitions that involve ferroelectric distortions and rotations of structural polyhedra. These types of phase transitions are not restricted to perovskites, and we will also discuss phases of silica that have similar phase transitions. We also met phase transitions involving rotations of molecules (C_{60} and N_2), and phase transitions involving disordering of positions of atoms, as in the fast-ion conducting phases such as AgI.

In this chapter we are primarily concerned with phase transitions in which the crystal structure of one phase is a distorted form of the crystal structure of the other phase, and where the transition is accomplished by small displacements of the atoms. These are called **displacive phase transitions**. This point is illustrated in Fig. 12.1, in which we show the crystal structures of the phases of the silica (SiO_2) polymorphs quartz and cristobalite. Both polymorphs undergo displacive phase transitions, in which the symmetry of one phase is lowered by small displacements of some of the atoms. In quartz, the change in symmetry

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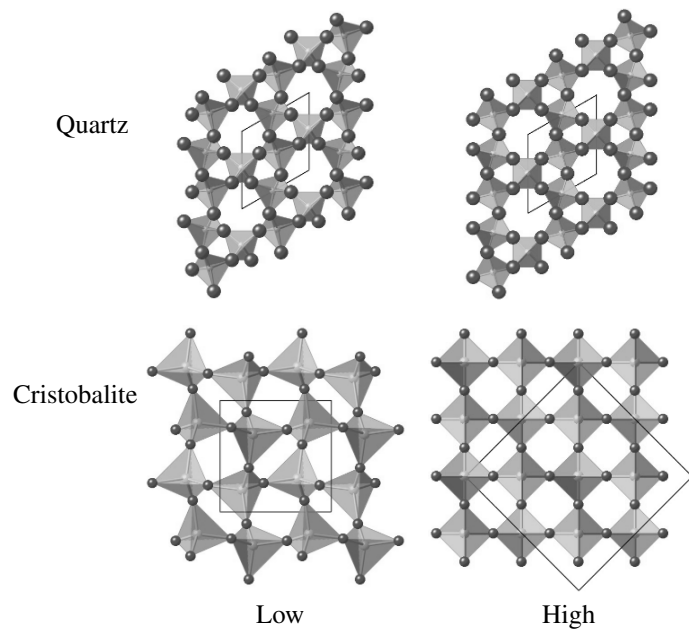


Fig. 12.1 Crystal structures of the two phases of quartz and cristobalite.

is produced by small rotations of the SiO_4 tetrahedra, as a result of which the 6-fold rotation symmetry is replaced by 3-fold rotation symmetry. Thus the symmetry of the lattice is lowered from hexagonal to trigonal, with a change in the point group from 622 to 32 (the space group changes from $P6_222$ to $P3_221$). In the case of cristobalite, there are similar rotations of the SiO_4 tetrahedra, and in this case the changes in symmetry are from face-centred cubic (point group $m\bar{3}m$, space group $Fd\bar{3}m$) to tetragonal (point group 422, space group $P4_12_12$), with the unit cell of the tetragonal phase encompassing two primitive unit cells of the cubic phase. The phase transitions in the perovskite structures we met earlier are other examples of displacive phase transitions.

The defining feature of a displacive phase transition may be understood to be a change in symmetry. There are cases of phase transitions where it appears that the change in symmetry is not a defining feature, but these cases are rare, and when reported they may need to be treated with some caution. The point about symmetry is that it is either present or absent, there is no intermediate state. The deformation of the structure may be reduced continuously on heating up to a phase transition, but whilst there is even an infinitesimal distortion of the structure, the symmetry is lost. The changes of symmetry associated with phase transitions in crystalline silica polymorphs have been highlighted above. In the case of perovskites, the ferroelectric distortion on transforming from the cubic phase of point group $m\bar{3}m$ to $4mm$ involves loss of the 3-fold axes and the centre of symmetry. It is important to note that the mere transition from cubic to tetragonal, as would be achieved by a simple expansion of the lattice along [001], does not describe the complete symmetry change; this is a point that will be important later when we discuss couplings between strain and atomic displacements. The rotational phase transitions in perovskites involve a

change in the description of the unit cell. The unit cell of the lower-symmetry phase encompasses two unit cells of the cubic phase in each direction. The change in symmetry in this case includes translational symmetry.

A phrase often used to describe the change of symmetry at a phase transition is **broken symmetry**. An example of broken symmetry is shown in Fig. 12.2. The medieval cathedral of Ely once had a symmetrical west end, with a large central tower and two side towers. The defining symmetry was a vertical mirror plane. Sadly, one tower collapsed and was never rebuilt. The loss of symmetry is quite clear.

The starting point to understanding the origin of phase transitions is with thermodynamics. We know that the equilibrium state of matter is that for which the free energy is a global minimum. This is an incredibly powerful statement. If we have some means to write down the free energy as a function of the critical variables (such as volume as a general variable, or some measure of the distortion of the structure as a specific variable), minimization of the free energy will yield an equation for the temperature or pressure dependence of these variables. It will also then become possible to obtain properties such as the susceptibility or heat capacity, quantities that are known to have an anomalous behaviour associated with a phase transition.



Fig. 12.2 A real case of broken symmetry: the mirror symmetry of the west front to Ely cathedral was lost when the north tower collapsed.

12.1.1 Importance of thermodynamic analysis

We recall from Chapter 5 that the free energy can be written as

$$G = U - TS + PV \quad (12.1)$$

It is nearly always the case that the lower-symmetry phase will be stable at lower temperatures and higher pressures, and the higher-symmetry phase will be stable at higher temperatures and lower pressures. We can rationalize this from the general form of the free energy in a way that shows the power of this concept. The internal energy U is primarily associated with the bond energy and the energy stored in the phonons. Clearly at low temperature the free energy will be dominated by the enthalpy, $H = U + PV$. Typically the distortion of the structure will lead to an increase in its density, and the lower volume will lead to a lower energy U and a lower value of PV , the latter term becoming more important at higher pressures. Hence we are able to understand why the distorted phase will be the stable phase at low pressure and low temperature. On heating, the entropy makes a growing contribution to the free energy. We can make a reasonable presupposition about the form of this contribution. The distorted phase has a more compact structure than the high-symmetry phase, and this will result in slightly higher phonon frequencies and hence a lower entropy (Section 9.1.5). Thus the $-TS$ term will favour the high-symmetry structure over the distorted one when T is sufficiently large.

In principle exact functions can be obtained for the different terms in the free energy, and these can be related to microscopic interactions. Later in this chapter (Section 12.5) we will derive this link, but first we describe several different types of displacive phase transitions.

12.1.2 Various types of displacive phase transitions

Ferroelectric phase transitions

In Section 2.4 we outlined the main features of ferroelectric phase transitions, namely that the symmetry-breaking displacements of the atoms allowed the formation of a dielectric polarization of the crystal. We also noted that the phase transition is accompanied by a divergence of the dielectric susceptibility.

One of the important characteristics of ferroelectric phase transitions is that the dielectric polarization can be reversed by the application of an electric field. This is not surprising, since the field only has to reverse the directions of the relatively small static atomic displacements. How this reversal takes place is illustrated in Fig. 12.3. The field is first applied along the direction of the polarization. This increases the polarization by an amount determined by the susceptibility, $\Delta P = \chi E$. The field is then reduced to zero, so that the polarization returns to its equilibrium value P_0 . The field is then increased in the opposite direction. The polarization is then reduced, instead of immediately switching direction. It is only when the reverse field reaches a certain strength, known as the **coercive field strength**, labelled E_c in Fig. 12.3, that the polarization suddenly switches directions. Then the polarization increases in magnitude along the opposite direction following the susceptibility. The reverse process then happens when the field is reduced and then turned on in the opposite direction. The resultant plot of polarization against electric field has the shape of a loop, which is called the **hysteresis loop**, as shown in Fig. 12.3. The same type of hysteresis loops are also found in ferromagnets, and it is the close analogy between the ferromagnetic and ferroelectric phase transitions that led to the naming of the latter.

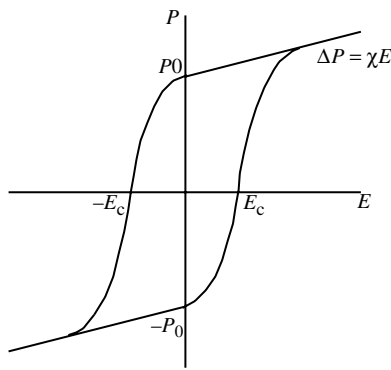


Fig. 12.3 Ferroelectric hysteresis loops.

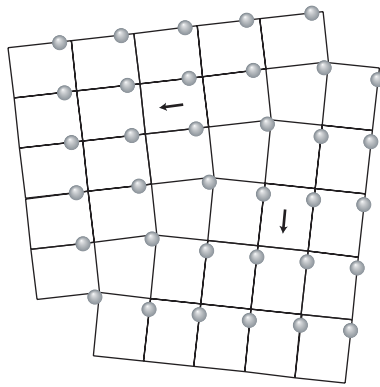


Fig. 12.4 Ferroelectric domain wall, showing motion of domain wall under application of an electric field.

In principle, the size of the coercive field could be related to the size of the potential energy barriers against movements of individual atoms, or of a free energy barrier. However, the most important factor concerns the mechanism by which the polarization can be reversed. When a material undergoes a ferroelectric phase transition, it can form small regions of the crystal in which the polarization points along different directions. In PbTiO_3 , the polarization can point along any of the $\langle 001 \rangle$ directions of the parent cubic phase, which means that there are six possible polarization states of the ferroelectric field. In practice, all six orientations can be found in different places in one crystal. These regions are called **domains**, and the boundaries between the regions are called **domain walls**. A schematic illustration of a domain wall is shown in Fig. 12.4. Switching the polarization direction of any domain will take place at the domain wall, since the atoms bounding the domain walls do not feel the effects of so many aligned neighbours. This is illustrated in Fig. 12.4. The effect of switching the polarization of the unit cells close to the domain wall has the appearance of the domain wall moving through the material. It is the force required to move the domain wall that determines the size of the coercive field, and clearly this may be affected by the presence of defects that can act to pin domain walls. Domains and domain walls are discussed in Appendix A.

Antiferroic (unit-cell combining) displacive phase transitions

In a ferromagnet, the magnetic moments on the atoms in each unit cell align in the same direction. However, there is a large class of magnetically-ordered

materials in which the magnetic moments of the atoms in adjacent unit cells point along opposite directions – a simple example of this ordering is shown in Fig. 12.5 (examples are oxides with the NaCl structure, such as NiO and MnO). These are called **antiferromagnetic** materials. There are many displacive phase transitions that are analogous to this, and these are called **antiferroelectric** phase transitions, if there are similarities to the displacements found in ferroelectric phase transitions, or just **antiferroic** displacive phase transitions in more general cases. The point is that there are many phase transitions in which displacements of the atoms in one unit cell are balanced by opposite displacements of the atoms in neighbouring unit cells. Thus the unit cell of the low-symmetry phase will be a combination of two, four or eight unit cells of the high-symmetry phase.

A simple example is the rotational phase transition in SrTiO_3 , which was discussed in Chapter 2 and shown in Fig. 12.6. The oxygen atoms are displaced in a way that corresponds to the rotations of the TiO_6 octahedra. The resultant change in symmetry does not allow for the development of a dielectric polarization. Most displacive phase transitions are of this type, including quartz and cristobalite shown in Fig. 12.1.

Antiferroic phases will also show formation of domains, and examples are given in Appendix A. The number of types of domains that can be formed will depend on symmetry.

Ferroelastic phase transitions

Most displacive phase transitions will involve a change in the size and shape of the unit cell. For example, as a result of the ferroelectric phase transition in the perovskite PbTiO_3 , the c axis will expand relative to the a and b axes as the lattice type changes from cubic to tetragonal. This strain deformation of the lattice is called **spontaneous strain**. In the case of the perovskite ferroelectric, the size and sign of the spontaneous strain are independent of whether the polarization points along $\langle 001 \rangle$ or in the opposite direction, $\langle 00\bar{1} \rangle$. This is because the structures of the two domains are simply mirror images of each other, and thus the symmetry requires the spontaneous strains to be identical in both cases. This is achieved if the relationship between the spontaneous strain and polarization does not involve the sign of the polarization, which means that, to lowest order, the spontaneous strain $\epsilon_s \propto P^2$. When generalized to any phase transition, we often find that ϵ_s is proportional to the square of the atomic displacements.

However, there is a class of phase transitions in which the spontaneous strain can change sign. These are called **ferroelastic phase transitions**. In a manner similar to ferromagnetic and ferroelectric phase transitions, the strain can be reversed by application of a suitable external stress. The analogy with the ferromagnetic and ferroelectric phase transitions also extends to the point that the quantity analogous to the magnetic or dielectric susceptibility, namely the elastic compliance (the inverse of the elastic constant), will diverge at a ferroelastic phase transition. This is seen as a softening of one or more of the elastic constants of the material.

One example of a ferroelastic phase transition is found in Na_2CO_3 , which is shown in Fig. 12.7. In this case the hexagonal high-temperature phase transforms to a monoclinic phase by a shear in the $\mathbf{a}$ – $\mathbf{b}$ plane. This shear is accompanied by the relative shifts of the columns of NaO_6 octahedra, which

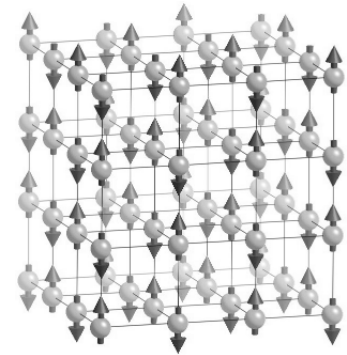


Fig. 12.5 Antiferromagnetic ordering in a simple cubic lattice.

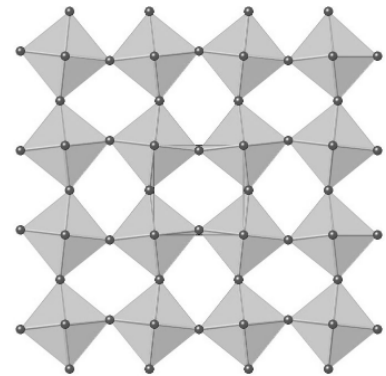
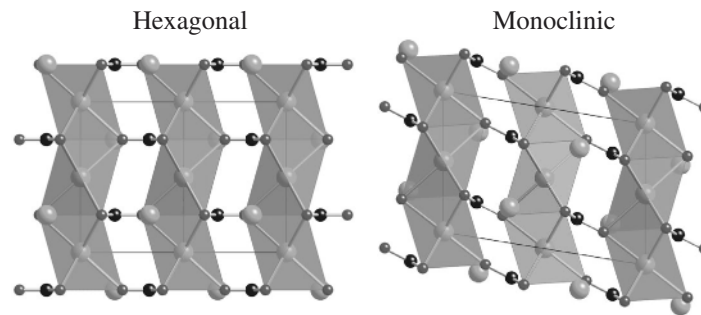


Fig. 12.6 Octahedra rotation phase transition in perovskite showing opposite rotations of octahedra leading to a doubling of the unit cell in each of the two directions in the plane of the diagram.

Fig. 12.7 Crystal structures of the two phases of Na_2CO_3 , showing how the low-temperature monoclinic phase can be obtained by a shear strain of the high-temperature hexagonal phase.



are hinged by the rotating planar carbonate molecular ions. The shear strain is proportional to the size of the rotation, and can be reversed by application of the corresponding shear stress.

The reader may permit the author to interject with some additional detail on the ferroelastic phase transition Na_2CO_3 . The shear instability involves the value of the c_{44} elastic constant in the hexagonal phase vanishing at the phase transition. Because of the hexagonal symmetry, $c_{55} = c_{44}$, and the shear elastic instability occurs for any direction in the **a**–**b** plane. It has been shown that this type of instability gives rise to a divergence of the atomic mean-square displacements, and these in turn cause the Bragg peaks with $l \neq 0$ to vanish at the phase transition. At this point, there is only true long-range order in the [001] direction, and the material has order that resembles a two-dimensional liquid in all directions normal to [001]. More details are given in Harris *et al.*, *J. Phys.: Cond. Matter* **8**, 7073, 1996, and references cited therein.

12.2 Quantitative description of displacive phase transitions: the concept of the order parameter

12.2.1 The general definition of the order parameter

The descriptions of the displacive phase transitions given above have been in terms of symmetry-breaking atomic displacements that give rise to a change in symmetry. For many systems there will be several atoms moving as a result of the phase transition. For example, in ferroelectric perovskites such as PbTiO_3 , the phase transition involves displacements of both cations and anions. When several atoms are involved in the symmetry-breaking deformations associated with the phase transition, it is usually possible to identify an overall pattern of relative atomic displacements, and it is the amplitude of the overall pattern of displacements that changes with temperature. In principle one might want to generate a thermodynamic theory that takes account of all symmetry-breaking displacements individually, but this could be rather complicated. Instead it is common to treat the pattern of atomic displacements as a fixed pattern, and the overall amplitude is the quantity that would be used in any theoretical treatment. This quantity is what is called the **order parameter**. This definition is analogous

to the concept of the normal mode coordinate in the theory of lattice dynamics. The pattern of displacements associated with any normal mode of the crystal is defined by the pattern of force constants, and is independent of the amplitude of the mode. The normal mode coordinate gives the overall amplitude of the displacement at any time, and this is the quantity that contains the dependence on temperature. Following the analogy further, the low-symmetry phase can be described as arising from a freezing of one of the normal modes of the high-symmetry phase into the structure as a static deformation, and in this case the order parameter is then the static equivalent of the normal mode coordinate. This suggests that the lattice dynamics of a crystal are implicated in the origin of a phase transitions, a point that we will develop later in this chapter (Section 12.4).

The power behind the use of the order parameter is that with a single parameter (or a small number of parameters in more complex cases) it becomes possible to develop a theory of the phase transition to any level of accuracy one wishes to aim at, and within any theoretical framework involving the order parameter it may also be possible to encompass a wide range of phenomena. Moreover, aside from the symmetry of the deformation, the exact pattern of atomic displacements may often be of little relevance in any subsequent theory. Continuing the analogy with the normal mode coordinate in the theory of lattice dynamics, the mode eigenvectors do not enter into the thermodynamics of the lattice vibrations. The only quantity that is of importance is the mode frequency, through which we obtain the normal mode coordinate and thence the thermodynamic functions. The point is that it should be possible to obtain an expansion of the thermodynamic free energy in terms of the order parameter. From the free energy it is possible to calculate any other function, and hence it ought to be possible to construct a theoretical framework that has a simple dependence on order parameter and which encompasses all experimental data. This would give a broad unifying description, and could be a predictive tool.

Before we consider some specific examples, it is worth developing a graphical illustration of the central point. In Fig. 12.8 we show three atoms in each unit cell that are involved in the symmetry-breaking distortion associated with a displacive phase transition. One of the atoms lies on a mirror plane, and as a

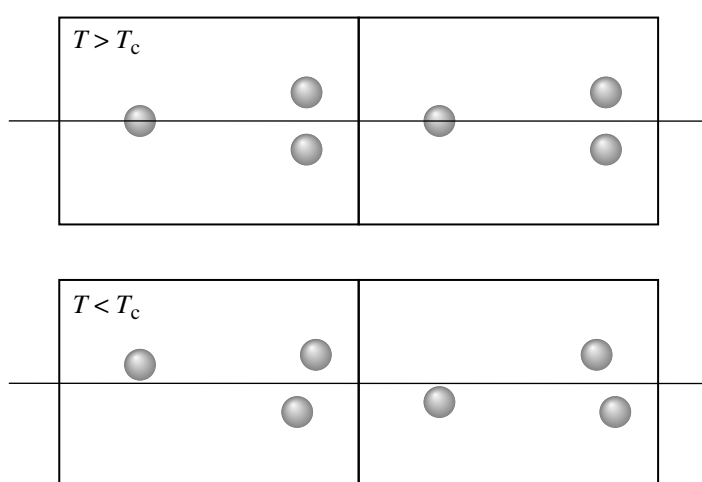


Fig. 12.8 Illustration of symmetry-breaking displacements. The mirror plane is converted to a glide plane as the lattice repeat doubles.

result of the phase transition the atoms in alternate unit cells move in opposite directions normal to the mirror plane. The other atoms are also displaced in order to break the mirror symmetry. There are two effects on the symmetry as a result of the phase transition. The first is the loss of the mirror plane, and the second is the doubling of the repeat distance along the direction of the mirror plane. These two symmetries are replaced by the glide symmetry. The order parameter describes how far the atoms move to break the symmetry – in effect, the order parameter gives a measure of how much the symmetry has been broken by. Of course, a change in symmetry is step-wise: the symmetry is either present or is not present. However, the changes that accompany the loss of symmetry will be gradual to some extent, and the order parameter gives a measure of these gradual changes. This is very similar to the discussion of symmetry-breaking displacements associated with normal modes of vibrations given in Sections 3.4 and 8.5.4.

We noted earlier that some phase transitions may actually require the use of more than one order parameter. One example is the ferroelectric phase transition in the perovskites, in which the polarization could point along any of the three axes. We denote these three components as P_x , P_y , and P_z . For the phase transition in PbTiO_3 , $P_y = P_z = 0$, and we only need to consider the variation of P_z . For BaTiO_3 , the first phase transition on cooling is similar to that in PbTiO_3 . The second phase transition then involves forming $P_y \neq 0$. Finally, all three components become active with $P_x = P_y = P_z \neq 0$. Similarly, the rotational phase transitions in the perovskites can be defined by three rotation angles R_x , R_y , and R_z . For SrTiO_3 , $R_y = R_z = 0$. However, there are some rotational phase transitions in which the behaviour is more complicated. In LaAlO_3 , there are equal rotations about all three axes, i.e. $R_x = R_y = R_z$. But in other perovskites, such as CaTiO_3 , there are unequal rotations about the three axes, $R_x \neq R_y \neq R_z \neq 0$. Complete descriptions of these phase transitions will require the use of all three order parameters.

12.2.2 Examples of order parameters for specific phase transitions

The concept of the order parameter is illustrated by several of the examples of phase transitions we have already encountered. In the octahedral tilting phase transition in SrTiO_3 the order parameter is simply the angle through which the TiO_6 octahedra rotate. These rotations involve displacements of oxygen atoms, and there is only one type of atomic displacement. The phase transitions in quartz and cristobalite involve rotations of SiO_4 tetrahedra as analogues to the octahedral-tilting phase transition in SrTiO_3 . However, these rotations are accompanied by displacements of the tetrahedra, and there is a more complex pattern of atomic displacements. Nevertheless, the pattern of displacements can be described by a single order parameter in each case.

In none of these examples is there a macroscopic analogue of the order parameter. On the other hand, the ferroelectric phase transitions do have a clear macroscopic equivalent of the order parameter. We recall that in PbTiO_3 there are displacements of cations and anions. The anions lying along the z direction from the Ti cation are allowed to have different displacements to the oxygens in the same x – y plane as the cations. The electrical polarization that is generated

by these displacements is given as

$$P = N \left[q(\text{Ti})\delta(\text{Ti}) + q(\text{Pb})\delta(\text{Pb}) + \sum_{j=1}^3 q(\text{O}_j)\delta(\text{O}_j) \right] \quad (12.2)$$

12.2.3 Order parameters in other phase transitions

We have illustrated the concept of the order parameter through a number of different displacive phase transitions. However, the concept of the order parameter actually transcends specific details of the type of phase transition, and can be used to describe any phase transition. The common example is a magnetic phase transition. A magnetic system will contain ions with magnetic moments. What changes as a result of the phase transition is not the displacements of the magnetic moments, but their orientations. In the high-temperature paramagnetic phase the orientations of the individual magnetic moments will be random (although they may be constrained to a few particular orientations). Thus the average moment of the crystal obtained by summing over all moments is zero. Below the phase transition, the orientations of all atomic moments become progressively ordered. These two situations are illustrated in Fig. 12.9. If we write the individual moments as $\mathbf{m}_j$, the order parameter can be written as

$$\langle \mathbf{m} \rangle = \frac{1}{N} \sum_j \mathbf{m}_j \quad (12.3)$$

In this case, the phase transition involves a real ordering of the orientations of the magnetic moment, and it is in this context that the name **order parameter** has its origin.

Similar ordering processes can be seen in phase transitions involving orientations of molecules. Examples from earlier in the book include the ordering of the orientations of N_2 and C_{60} molecules. The same idea will apply to the ordering of orientations of long-chain molecules in some liquid crystals.

Another example of an ordering process occurs in alloys such as Cu_3Au . In the high-temperature phase, the Cu and Au atoms are distributed randomly across all sites in a ccp structure, but on cooling below a particular temperature the Cu and Au atoms begin to occupy certain preferred sites, until at low temperature there is complete ordering of the positions of the different atoms in a tetragonal unit cell. We use x_{Cu} to represent the occupancy of the low-temperature Cu sites by Cu atoms. At low temperature $x_{\text{Cu}} = 1$, and in the high-temperature disordered phase $x_{\text{Cu}} = 3/4$. We can therefore quantify the degree of order by the order parameter

$$4 \left(x_{\text{Cu}} - \frac{3}{4} \right)$$

This type of site-ordering phase transition is found in many system where there are cations that are able to occupy identical sites without too high a cost in energy.

12.2.4 Experimental measurements of order parameter

For ferroelectric phase transitions, the dielectric polarization is a macroscopic representation of the order parameter, and can be measured using macroscopic

$T > T_c ; \eta = 0$



$T < T_c ; \eta > 0$



Fig. 12.9 Illustration of the calculation of the order parameter for the ordering of an array of spins.

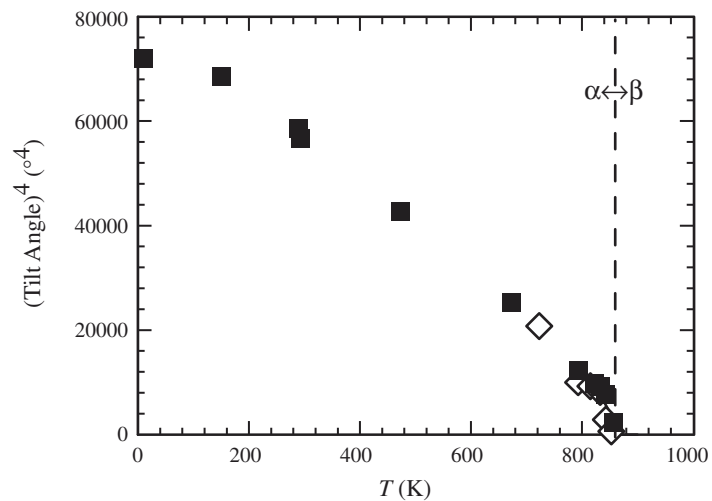


Fig. 12.10 Temperature dependence of the fourth power of the angle of rotation of SiO_4 tetrahedra associated with a displacive phase transition in quartz – the fourth power is plotted because the temperature dependence is almost linear except for the first-order discontinuity at the phase transition temperature.

techniques. However, in most cases there will be no macroscopic representation of the order parameter, and then it will be necessary to perform a direct measurement of the atomic displacements through measurements of the crystal structure, or else obtain information about the order parameter through indirect measurements.

Figure 12.10 shows the temperature dependence of the angle through which the SiO_4 tetrahedra rotate following the displacive phase transition in quartz (the rotation is shown in Fig. 12.1). The figure combines both X-ray and neutron diffraction data (Chapter 6).

Indirect measurements of the order parameter require that some measurable property of the material will change as a result of the phase transition, and the change can be related to the value of the order parameter. The value of an indirect measurement of the order parameter is that it can often be obtained with higher precision than through direct diffraction measurements. Examples of indirect measurements are intensities of Bragg peaks that vanish on heating into the high-symmetry phase (see Problem 12.12), changes in the lattice parameters that accompany the phase transitions, or changes in phonon frequencies measured by infrared or Raman spectroscopy (Chapter 10; Section 12.5.1). Often, as in measurements of changes in vibrational frequencies, the change will be independent of the sign of the order parameter, and hence will be proportional to the square of the order parameter.

In Fig. 12.11 we show measurements of the piezoelectric coefficients of quartz. One of the coefficients becomes zero in the high-temperature phase, and by symmetry its variation with temperature below the phase transition mirrors the temperature dependence of the order parameter.

12.2.5 First- and second-order phase transitions

We have met examples of the temperature dependence of the order parameter and quantities that depend on the order parameter. In some cases the order parameter falls continuously to zero at the transition temperature T_c , and in

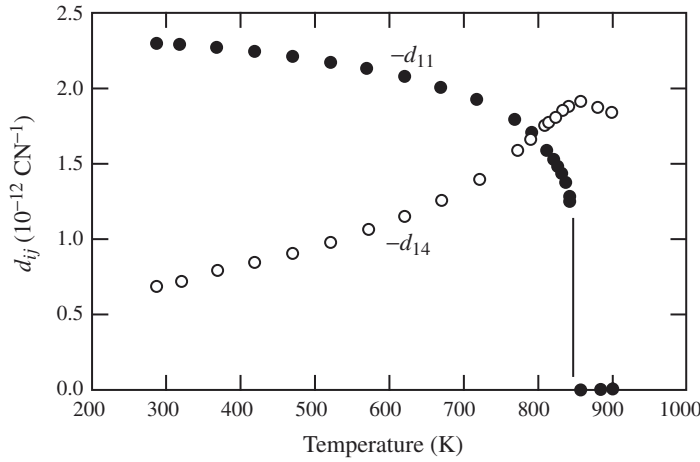


Fig. 12.11 Temperature dependence of the piezoelectric coefficients of quartz.

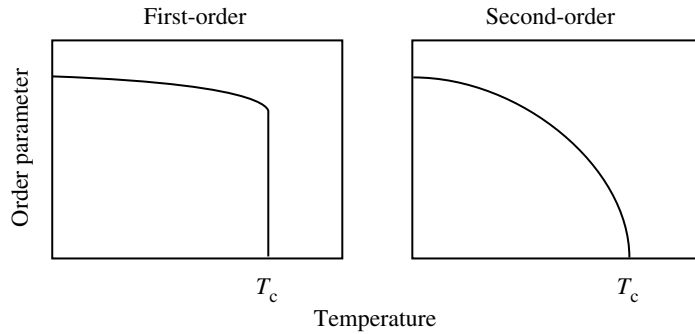


Fig. 12.12 Schematic representation of temperature dependence of order parameters at first and second order phase transitions.

other cases there is a discontinuous change at the transition temperature. These differences define two types of phase transitions: the case where there is a discontinuous change is called a **first-order** phase transition, and the case where the change is continuous is called a **second-order** phase transition. The names come from an old characterization, where the order of a phase transition was associated with the lowest-order derivative of the free energy with temperature that changes discontinuously at the phase transition. Thus at a first-order phase transition, the entropy $S = -\partial F/\partial T$ changes discontinuously at the transition temperature, and at a second-order phase transition the heat capacity $c = T\partial S/\partial T = -T\partial^2 F/\partial T^2$ changes discontinuously. In principle one might imagine the possibility of a third-order phase transition, but this concept is not useful since it represents a special case that can be understood within the concept of second-order phase transitions. The ideas can be generalized for pressure-induced displacive phase transitions. The two types of phase transitions are illustrated in Fig. 12.12.

In many second-order phase transitions, the order parameter, which we label as η (this is often given the symbol Q , but this is liable to get confused with the symbol for normal mode coordinate), varies as

$$\eta \propto |T_c - T|^{1/2} \quad (12.4)$$

In this case, any property that varies as η^2 will fall linearly to zero at the phase transition. In first-order phase transitions, the analytical form of η is more complex (to be discussed below; Section 12.3.4). Some phase transitions seem to be almost exactly part-way between second and first order, and in these cases the order parameter varies as

$$\eta \propto |T_c - T|^{1/4} \quad (12.5)$$

Such cases are called **tricritical phase transitions**, because in some cases this behaviour is found close to a tricritical point in a phase diagram. Quartz is an example of a phase transition that is strictly first order, but the discontinuity is so small that for a wide range of temperatures the order parameter varies in a way that is close to that of a tricritical phase transition. The temperature dependence of the SiO_4 rotation angle in quartz in Fig. 12.10 is plotted in a way that highlights this point.

12.3 Landau theory of displacive phase transitions

12.3.1 Qualitative behaviour of the free energy

Displacive phase transitions lend themselves to a reasonably accurate general free energy analysis. In this section we will develop an empirical but quantitative thermodynamic description of phase transitions within the context of what is known as **Landau theory**. Later in this chapter (Section 12.5) we will show that the basic ideas can be justified from a lattice dynamics perspective.

As we have constantly stressed, the starting point has to be the free energy, since this contains information about the equilibrium structure. In this chapter we will use the Helmholtz free energy. Consider a material at a temperature above that of a second-order phase transition. Its free energy as a function of η will have a single minimum at $\eta = 0$, as shown in Fig. 12.13.

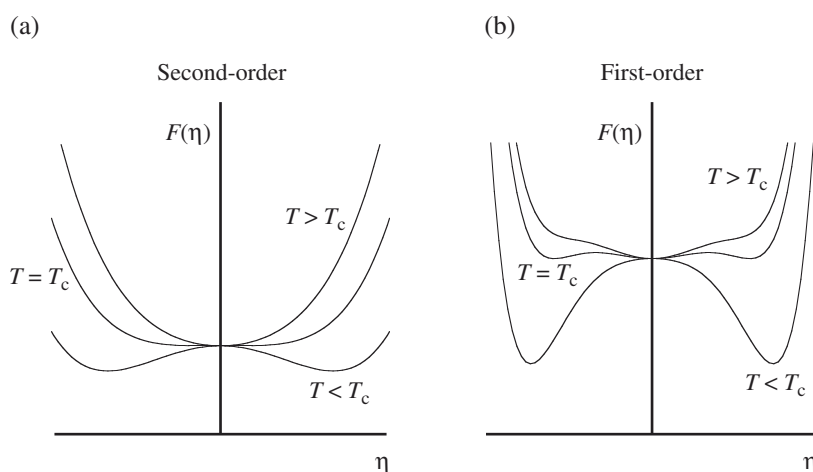


Fig. 12.13 Variation of free energy curves with order parameter for second-order and first-order phase transitions and temperatures below, at and above the transition temperature.

At temperatures far below the transition temperature, the free energy curve, $F(\eta)$, will have a global minimum at non-zero values of η . In a second-order phase transition, the position of this minimum gradually moves closer to $\eta = 0$ on heating towards T_c , as shown in Fig. 12.13. For temperatures above T_c there is only a single minimum at $\eta = 0$.

At a first-order phase transition, the picture is more complicated. At the transition temperature, there must be minima at both $\eta = 0$ and at non-zero values of η . The overall variation of the free energy curves for a first-order phase transition is shown in Fig. 12.13.

12.3.2 Expansion of the free energy function for a second-order phase transition

At high temperatures, the free energy curves have a single minimum, and we can express the free energy function as

$$F(\eta) = F_0 + \frac{1}{2}A\eta^2 + \dots \quad (12.6)$$

where A is a positive coefficient. F_0 is the part of the free energy that does not change through the phase transition, and we will not always include this in the subsequent equations. At the transition temperature, the free energy function at $\eta = 0$ switches from being a minimum to a maximum. This corresponds to the coefficient A changing sign at $T = T_c$. Since the process is gradual, one simple representation of the temperature dependence of A is

$$A(T) = a(T - T_c) \quad (12.7)$$

Since the free energy function curves downwards around $\eta = 0$ where $T < T_c$, the function must contain an additional positive term with a higher-order power of η in order that to $F(\eta)$ will curve upwards again at larger values of η . The lowest-order term that we can use is the quartic term, so we can now expand the free energy function as

$$F(\eta) = F_0 + \frac{1}{2}a(T - T_c)\eta^2 + \frac{1}{4}b\eta^4 \quad (12.8)$$

The development of this form of the free energy function has been intuitive, but it is worth rehearsing the principles behind the intuition. We have used the constraints on the form of the free energy to ensure that it gives the correct minima across a range of temperatures. We have also used symmetry constraints, namely that $F(\eta) = F(-\eta)$, to demonstrate that the free energy has no odd powers of η . We have also used the property that F varies continuously through the phase transition, so that all parameters must be continuous functions of temperature.

12.3.3 Calculation of properties for a second-order phase transition

The principle that the equilibrium state is that for which the free energy is a minimum gives us a method for calculating the temperature dependence of the

properties. The equilibrium value of the order parameter for any temperature is that given by the solution of the equations

$$\frac{\partial F}{\partial \eta} = 0 \quad \frac{\partial^2 F}{\partial \eta^2} > 0 \quad (12.9)$$

The first differential gives

$$a(T - T_c)\eta + b\eta^3 = 0 \quad (12.10)$$

There are three solutions in principle. One is $\eta = 0$. When $T > T_c$, this solution corresponds to the minimum of the free energy, and there are no other real solutions. When $T < T_c$, the solution $\eta = 0$ corresponds to a maximum point of the free energy, and the other two solutions now correspond to the minima in the free energy function:

$$\eta = \pm \sqrt{\frac{a}{b}} |T_c - T|^{1/2} \quad (12.11)$$

This describes the temperature dependence of the order parameter found in many phase transitions.

We now unpack the free energy to obtain its separate contributions:

$$\begin{aligned} F - F_0 &= \Delta U - T \Delta S \\ \Rightarrow \Delta U &= -\frac{1}{2}aT_c\eta^2 + \frac{1}{4}\eta^4 \end{aligned} \quad (12.12)$$

$$\Delta S = -\frac{1}{2}a\eta^2 \quad (12.13)$$

Thus the internal energy, which is assumed to be independent of temperature, is a double-well function. The entropy decreases as a quadratic function of the order parameter, which we will understand as arising from the increase in phonon frequencies as the structure is distorted.

The contribution to the heat capacity from the phase transition is obtained as

$$\Delta c(T < T_c) = T \frac{\partial S}{\partial T} = -\frac{1}{2}aT \frac{\partial \eta^2}{\partial T} = \frac{a^2 T}{2b} \quad (12.14)$$

$$\Delta c(T > T_c) = 0 \quad (12.15)$$

Thus the total heat capacity shows a small step change at the second-order phase transition temperature.

The susceptibility can be obtained from the second-order differential of the free energy with respect to the order parameter (Problem 12.1):

$$\chi^{-1} = \frac{\partial^2 F}{\partial \eta^2} = a(T - T_c) + b\eta^2 \quad (12.16)$$

When $T > T_c$, $\eta = 0$ and

$$\chi^{-1} = a(T - T_c) \quad (12.17)$$

On the other hand, when $T < T_c$, substitution of the equation for $\eta(T)$ (eqn 12.11) yields

$$\chi^{-1} = 2a(T_c - T) \quad (12.18)$$

Clearly $\chi^{-1} \rightarrow 0$ as $T \rightarrow T_c$, which corresponds to a divergence of the susceptibility at the phase transition. In a ferroelectric phase transition, this accounts for the divergence of the dielectric constant as seen in Figs 1.8 and 2.31.

12.3.4 First-order phase transitions

At a second-order displacive phase transition, the free energy function at $\eta = 0$ changes from being a minimum to a maximum at the phase transition. However, at a first-order phase transition, a minimum at $\eta = 0$ will coexist with a second minimum at a non-zero value of η . We will restrict our analysis to the case where $F(\eta) = F(-\eta)$, so there will be three coexisting minima at the phase transition, but there are cases where this symmetry is not required. The symmetric case is shown in Fig. 12.13. This is achieved by a small modification of the Landau free energy function for the second-order phase transition:

$$F(\eta) = \frac{1}{2}a(T - T_0)\eta^2 - \frac{1}{4}b\eta^4 + \frac{1}{6}c\eta^6 \quad (12.19)$$

where we use T_0 rather than T_c because T_0 is not to be identified with the transition temperature. The main difference from the free energy function for the second-order phase transition is that the fourth-order term is now negative (b is a positive number), and in order to ensure that the free energy function rises to large positive values for large values of $|\eta|$ it is necessary to have the positive sixth-order term.

The temperature dependence of the order parameter can be obtained by minimization of the free energy, yielding (see Problem 12.2)

$$\eta^2 = \frac{b + \sqrt{b^2 - 4ca(T - T_0)}}{2c} \quad (12.20)$$

The transition temperature is given as the temperature at which values of the free energy at each minimum are zero (Problem 12.2):

$$T_c = T_0 + \frac{3b^2}{16ac} \quad (12.21)$$

At this temperature the order parameter jumps from zero to $\pm(3b/4c)^{1/2}$. The change in internal energy at this temperature is

$$\Delta U = -\frac{3abT_c}{8c} = \frac{aT_c\eta^2}{2} \quad (12.22)$$

This is the latent heat for the phase transition.

There may be several reasons why a phase transition is first order. One is associated with the coupling to the spontaneous strain. We could expand the free energy as

$$F(\eta) = \frac{1}{2}a(T - T_0)\eta^2 + \frac{1}{4}b\eta^4 + \frac{1}{2}\lambda\epsilon_s\eta^2 + \frac{1}{2}C_{el}\epsilon_s^2 + \dots \quad (12.23)$$

where ϵ_s is a spontaneous strain produced by the phase transition, C_{el} is the corresponding elastic constant, and λ is another constant. The term of the form $\epsilon_s\eta^2$ has the appropriate symmetry for a phase transition that is not ferroelastic, which is seen by solving for the equilibrium condition:

$$\frac{\partial F}{\partial \epsilon_s} = 0 \Rightarrow \epsilon_s = -\frac{\lambda\eta^2}{2C_{el}} \quad (12.24)$$

(see Section 12.1.2.) Substituting into eqn 12.23 gives

$$F(\eta) = \frac{1}{2}a(T - T_0)\eta^2 + \frac{1}{4}\left(b - \frac{\lambda^2}{2C_{\text{el}}}\right)\eta^4 + \dots \quad (12.25)$$

The important point is that the coupling to strain has reduced the coefficient of the fourth-order term, and if the coupling were sufficiently large this coefficient could become negative. In this case, the phase transition would become first-order (and we would then need to incorporate the sixth-order term in the expansion of F).

12.3.5 The range of validity of Landau theory

Landau theory provides a very powerful description of the thermodynamics of phase transitions. This power is derived from the fact that the free energy can be written in terms of the order parameter, which in turn allows the free energy to be written in terms that encompass other quantities (such as elastic strain).

Although Landau theory is rooted in thermodynamics, the Landau free energy function $F(\eta)$ is not itself a true thermodynamic function. This is best indicated by the low-temperature behaviour. For a second-order phase transition the temperature dependence of η has the same form for all temperatures. More precisely, the derivative of the order parameter with respect to temperature is obtained from

$$2\eta \frac{\partial \eta}{\partial T} = -\frac{2a}{b} \quad (12.26)$$

In the limit $T \rightarrow 0$ this result contrasts sharply with the usual thermodynamic limiting behaviour $\partial \eta / \partial T|_{T \rightarrow 0} = 0$. This is obtained by starting from the basic definition of the internal energy

$$dU = T dS + H d\eta \quad (12.27)$$

where H is a field conjugate to η , which need not be possible to produce in the laboratory (not to be confused here with the enthalpy). The differential form of the internal energy gives

$$T = \left(\frac{dU}{dS}\right)_{\eta}; \quad H = \left(\frac{dU}{d\eta}\right)_S \quad (12.28)$$

These then yield

$$\frac{dT}{d\eta} = \frac{d^2U}{dS d\eta} = \frac{dH}{dS} \quad (12.29)$$

This pair of equations implies that

$$\frac{d\eta}{dT} = \frac{dS}{dH} \quad (12.30)$$

Since $S = 0$ for all values of the conjugate field H at $T = 0$, it follows that, at $T = 0$,

$$\left.\frac{d\eta}{dT}\right|_{T=0} = 0 \quad (12.31)$$

This contrasts with eqn (12.26), which gives a non-zero gradient for $\eta(T)$ at $T = 0$. The point is that the formulation of Landau theory is clearly not appropriate for low temperature. We will understand this point better in Section 12.5.

12.4 Soft mode theory of displacive phase transitions

12.4.1 Basic idea of the soft mode

If we perform a lattice dynamics calculation on a high-temperature phase of a material that undergoes a displacive phase transition, at least one of the harmonic values of ω^2 will be calculated as negative (such a calculation corresponds to $T = 0$). This is because the crystal is at a potential energy maximum with respect to the distortions that accompany the phase transition. These distortions will correspond to the atomic displacements associated with this particular phonon. The negative value of ω^2 is a signature of a displacive instability. Such a calculation would correspond to a temperature of zero, leading to the conclusion that the stable phase at low temperatures is the lower-symmetry distorted phase. However, at high temperatures the high-symmetry phase becomes stable, and we therefore expect that this phonon will now have a positive value of ω^2 . This line of reasoning suggests that ω^2 has an intrinsic temperature dependence of the form sketched in Fig. 12.14, increasing in value (i.e. becoming less negative) on heating. Eventually the value of ω^2 will reach zero, and for temperatures above this point the high-symmetry phase will be stable. The temperature at which ω^2 reaches zero is the transition temperature for a second-order phase transition.

It is instructive to turn the picture round. We then have a high-temperature phase that has one phonon whose frequency falls on cooling, eventually reaching zero, at which point the material undergoes a displacive phase transition to a lower-symmetry structure. The phonon frequency is said to **soften**, because its value is determined by restoring forces that appear to soften on cooling. This phonon is called the **soft mode**. The atomic motions associated with the soft mode correspond closely to the way the structure distorts through the phase transition – one can think of the lower-symmetry distorted structure in terms of the symmetric structure modulated by a frozen-in phonon distortion (recall the discussion in Section 12.1.2). For example, in the ferroelectric perovskites the soft mode involves the cations moving in opposite directions to the anions.

From the theory of the anharmonic phonon developed in Chapter 11 we can gain an understanding of the origin of the temperature dependence of the soft mode. Specifically we recall that the intrinsic temperature dependence of a phonon frequency due to direct anharmonic interactions (eqn 11.30) is given by

$$\omega^2 = \omega_0^2 + \frac{k_B T}{2} \sum_{\mathbf{k}', v'} \alpha_{\mathbf{k}', v'}^{(4)} / \omega^2(\mathbf{k}', v') \quad (12.32)$$

ω_0^2 is the negative harmonic value that we would calculate in the lattice dynamics calculation, and $\alpha_{\mathbf{k}', v'}^{(4)}$ is the coefficient of the fourth-order anharmonic

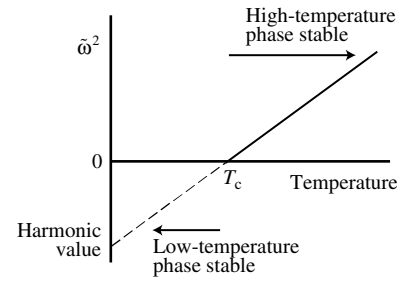


Fig. 12.14 Schematic representation of the temperature dependence of the square of the frequency of a soft mode. Below the temperature T_c the frequency is imaginary and hence unstable. The frequency at $T = 0$ K is the harmonic value.

interaction that couples the soft mode to a phonon of wave vector $\mathbf{k}'$, mode label ν' , and angular frequency $\omega(\mathbf{k}', \nu')$. Clearly, if sufficient values of $\alpha_{\mathbf{k}', \nu'}^{(4)}$ are positive, the effect of increasing temperature is to increase the value of ω^2 until it becomes positive. The phase transition occurs at the temperature at which $\omega^2 = 0$:

$$k_B T_c = \frac{-2\omega_0^2}{\sum_{\mathbf{k}', \nu'} \alpha_{\mathbf{k}', \nu'}^{(4)} / \omega^2(\mathbf{k}', \nu')} \quad (12.33)$$

If we now make use of the Einstein model, we can define an average angular frequency $\langle \omega \rangle$ and an average coupling coefficient $\langle \alpha^{(4)} \rangle$ (the averages can be weighted in order to ensure consistency), and therefore write

$$RT_c = \frac{-2\omega_0^2 \langle \omega \rangle^2}{3 \langle \alpha^{(4)} \rangle} \quad (12.34)$$

The temperature dependence of the soft mode is then given as

$$\omega^2 = \frac{|\omega_0^2|}{T_c} (T - T_c) \quad (12.35)$$

12.4.2 Ferroelectric soft modes

The soft mode theory of phase transitions was originally developed to explain the origins and mechanisms of ferroelectric phase transitions. In fact, the idea of the soft mode for ferroelectric phase transitions is implicit in the Lyddane–Sachs–Teller relation introduced in Section 8.5.6 to explain why the longitudinal (LO) and transverse (TO) optic modes at $\mathbf{k} = 0$ in ionic materials are not equivalent. We recall the main result (eqn 8.61):

$$\frac{\epsilon(\omega = 0)}{\epsilon(\omega = \infty)} = \frac{\omega_{\text{LO}}^2}{\omega_{\text{TO}}^2} \quad (12.36)$$

The point is that since the static dielectric constant, $\epsilon(\omega = 0)$, diverges at the ferroelectric phase transition, the LST relation implies that $\omega_{\text{TO}}^2 \rightarrow 0$ at the

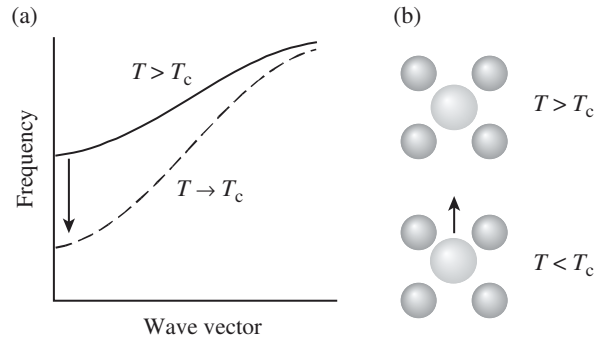


Fig. 12.15 Schematic representations of ferroelectric soft mode behaviour: (a) behaviour of the phonon dispersion curves; (b) atomic displacements.

same time. The softening of the transverse optic phonon branch is illustrated in Fig. 12.15, together with a schematic illustration of the atomic displacements associated with the soft mode.

The temperature dependence of the frequency of the soft mode associated with the ferroelectric phase transition in PbTiO_3 is shown in Fig. 12.16. The frequencies were measured using both inelastic neutron scattering (high-temperature phase) and Raman spectroscopy (low-temperature phase). At temperatures below the transition temperature, the soft mode hardens again. The two frequency values are for oscillations of the atoms in directions parallel to and normal to the direction of the dielectric polarization. The phase transition in PbTiO_3 is first order, so the soft mode frequency does not actually reach zero.

There are also phase transitions that have soft modes with zero wave vector but which are not ferroelectric. One example is quartz, which undergoes a hexagonal–trigonal phase transition at 846 K. Although both the high and low temperature phases are non-centrosymmetric, there is no accompanying change in the dielectric polarization or a divergence of the dielectric constant. The soft phonon mode involves the rotations of the SiO_4 tetrahedra shown in Fig. 12.1.

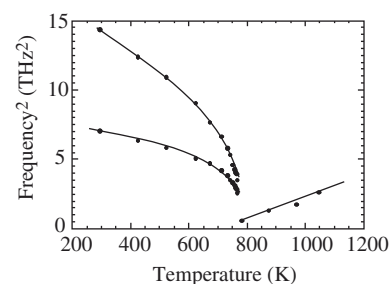


Fig. 12.16 Temperature dependence of the ferroelectric soft mode in PbTiO_3 . Data in the low-temperature phase are taken from the Raman scattering study of Burns and Scott (*Phys. Rev. Lett.* **25**, 167, 1970), and data in the high-temperature phase are taken from the inelastic neutron scattering study of Shirane *et al.* (*Phys. Rev. B* **2**, 155, 1970).

12.4.3 Zone boundary (antiferroelectric) phase transitions

A number of crystals undergo phase transitions which involve soft modes with wave vectors at Brillouin zone boundaries. In these cases the soft phonons can be either acoustic or optic modes. Because of mixing of eigenvectors, it is often the case that the distinction between the two types of mode at zone boundary wave vectors is not clear-cut (as pointed out in Chapter 8). The different types of soft mode, and the atomic displacements, are shown schematically in Fig. 12.17. One of the results of a zone boundary soft mode phase transition is that the unit cell of the low-temperature phase is doubled in one or more directions. In some cases neighbouring unit cells of the high-temperature form develop dipole moments, but as these are in opposite directions the unit cell at low temperature has no net moment.

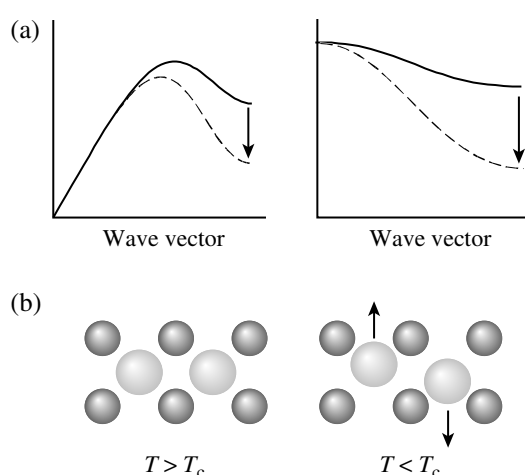


Fig. 12.17 Schematic behaviour of (a) zone boundary acoustic and optic soft modes; (b) atomic displacements showing doubling of the unit cell and cancelling induced dipole moments.

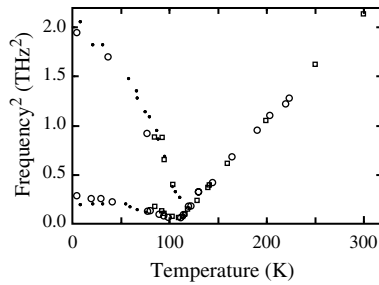


Fig. 12.18 The temperature dependence of the soft mode in SrTiO_3 , as measured by a combination of inelastic neutron scattering (open circles and squares, data from Sirane and Yamada, *Phys. Rev.* **177**, 858, 1969, and Cowley *et al.*, *Sol. State Comm.* **7**, 181, 1969, respectively) and Raman spectroscopy (closed circles, data from Fleury *et al.*, *Phys. Rev. Lett.*, **21**, 16, 1968).

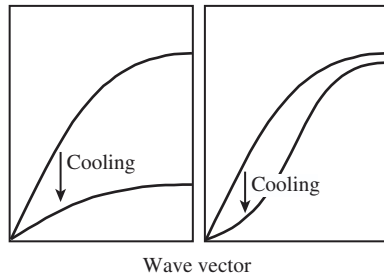


Fig. 12.19 Schematic representation of the softening of the acoustic mode. The left-hand figure shows the case where the acoustic mode softens across the whole branch. The right-hand figure shows the case where the softening is restricted to small wave vectors. In both cases, the softening leads to a reduction in the slope of the acoustic mode dispersion curve in the limit $k \rightarrow 0$.

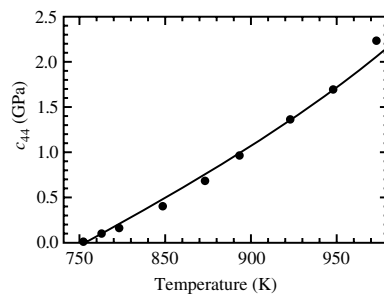


Fig. 12.20 Temperature dependence of the soft c_{44} elastic constant associated with the ferroelastic phase transition in Na_2CO_3 measured by inelastic neutron scattering. The data have been adapted from Harris *et al.*, *Phys. Rev. Lett.*, **79**, 4846, 1997.

Undoubtedly the best example of a zone boundary phase transition is the cubic–tetragonal transition in the perovskite SrTiO_3 (transition temperature 110 K). The soft mode involves the rotations of the interconnected TiO_6 octahedra about $[001]$, as shown in Fig. 12.6, with neighbouring octahedra in the (001) plane rotating in opposite directions. This distortion leads to a doubling of the repeat distance in all three directions, although this gives a face-centred tetragonal unit cell that can be reduced to a body-centred tetragonal unit cell by a 45° rotation of the axes about $[001]$. The soft mode has wave vector $[1/2, 1/2, 1/2]$. The soft mode in SrTiO_3 has been measured by both neutron scattering and Raman spectroscopy, and its temperature dependence is shown in Fig. 12.18.

12.4.4 Ferroelastic phase transitions

The soft modes considered above have been optic modes, or of optic character. Acoustic modes at $k = 0$ can also soften, but here the basic picture will be somewhat different since at $k = 0$ the frequency is already zero. In this case, the softening involves the slope of the acoustic mode falling to zero, and this corresponds to a softening of the elastic constant. Different ways in which this can happen are shown in Fig. 12.19. The phase transitions caused by a softening of an acoustic mode are the ferroelastic phase transitions discussed earlier in this chapter. The gradient of the soft acoustic mode as $k \rightarrow 0$ gives the velocity of sound, the square of which gives one of the elastic constants. The temperature dependence of the soft c_{44} elastic constant in ferroelastic Na_2CO_3 (see Section 12.1.2) has been determined from inelastic neutron scattering measurements of the acoustic modes, and the results are shown in Fig. 12.20.

12.4.5 Incommensurate phase transitions

Most soft-mode phase transitions involve modes with wave vectors at $k = 0$ or at one of the Brillouin zone boundaries. However, it is possible for the instability to occur at a wave vector that is at a more general position in the Brillouin zone. A schematic representation of the modulation of a crystal due to a frozen-in normal mode with a general wave vector is given in Fig. 12.21.

This type of phase transition is the **incommensurate phase transition** introduced in Chapter 2, and is so called because the wavelength of the modulation associated with the soft mode is incommensurate with the underlying crystal repeat distance (in metals the wave vector of the incommensurate soft mode will be associated with wave vector of the Fermi surface, Section 5.4.7). One example of an incommensurate phase transition is quartz: the famous α – β phase transition actually has an incommensurate phase existing over a small temperature range between the two better-known phases. The mechanism for this involves a softening of an optic mode and an anti-crossing interaction with an acoustic mode (anti-crossing was described in Section 8.5.7). The effects of the anti-crossing increase with k , since the optic and acoustic modes have different symmetry at $k = 0$, so the result is that the frequency of the acoustic mode is driven to zero at a non-zero value of k . This is illustrated in Fig. 12.22.

12.5 Lattice dynamical theory of the low-temperature phase

12.5.1 Lattice dynamical theories

The theory of the soft mode developed so far is strictly applicable to the dynamics of the high-temperature phase. From the theory it is possible to understand how the anharmonic interactions stabilize the soft mode, and the theory has given an equation for the transition temperature. Because the theory is only applicable to the high-temperature phase, it does not provide information about the temperature dependence of the order parameter. This is best obtained using a lattice dynamic model of the free energy, which we describe in this section. In this case, the soft mode becomes a static distortion of the crystal – we note that the order parameter η is now firmly identified with the static equivalent of the normal mode coordinate of the soft mode. In this case the potential energy part of the Hamiltonian has the form

$$\begin{aligned}\mathcal{H} &= \frac{1}{2} \sum_k \omega_k^2 Q_k Q_{-k} + \frac{1}{4} \alpha_k Q_k Q_{-k} \eta^2 \\ &= \frac{1}{2} \sum_k \left(\omega_k^2 + \frac{1}{2} \alpha_k \eta^2 \right) Q_k Q_{-k}\end{aligned}\quad (12.37)$$

In this equation k denotes all phonon modes excluding the soft mode, and the coefficient α_k denotes the anharmonic coupling of the order parameter with any phonon. This is equivalent to the coefficient $\alpha_{\mathbf{k}_0, -\mathbf{k}_0, \mathbf{k}, -\mathbf{k}}^{(4)}$, where we denote $\mathbf{k}, \nu$ by k , and $\mathbf{k}_0, \nu_0$ are the wave vector and branch label of the soft mode. The new Hamiltonian indicates that when there is a static distortion, the phonon frequencies are modified to give

$$\tilde{\omega}_k^2 = \omega_k^2 + \frac{1}{2} \alpha_k \eta^2 \quad (12.38)$$

If α_k is positive, the frequency is increased on cooling into the low-temperature phase, and this will have the effect of decreasing the phonon entropy as required. We also remark that the dependence of the phonon frequency on the order parameter given in this relationship can be measured in a spectroscopy experiment, giving an indirect method to determine the temperature dependence of η (see Section 12.2.4).

12.5.2 Potential energy of the crystal

Before we consider the phonons further, we also need to account for the change in potential energy of the crystal due to the phase transition. We write this as

$$E(\eta) = -\frac{1}{2} \kappa_2 \eta^2 + \frac{1}{4} \kappa_4 \eta^4 \quad (12.39)$$

This form has been anticipated by eqn 12.12, which gives the internal energy contribution to the Landau free energy function. Since we can identify the order parameter η as the static component of the frozen-in normal mode that is the

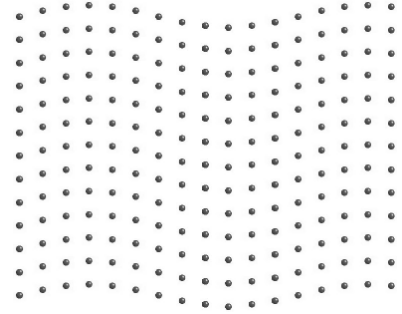


Fig. 12.21 Schematic representation of the modulation of a lattice caused by a soft mode with a general wave vector.

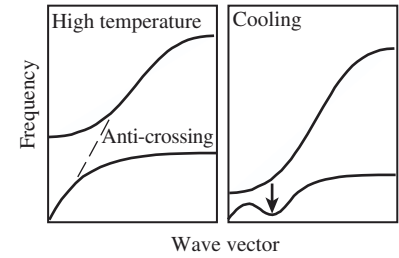


Fig. 12.22 Schematic representation of the mechanism of the incommensurate softening of an acoustic mode in quartz as a result of the softening of an optic branch.

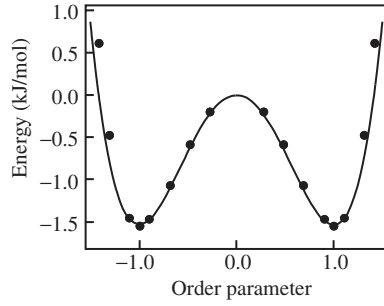


Fig. 12.23 Variation of the potential energy of quartz with changing order parameter. The points are energies calculated using a pair potential model as described in Chapter 5, and the curve is described by eqn 12.39 with fitted values of κ_2 and κ_4 . (The calculation was originally reported by Dove *et al.*, *Phys. Chem. Min.*, **26**, 344, 1999.)

soft mode for the phase transition, κ_2 can be identified with the soft mode ω^2 . Of course, η can be scaled by an arbitrary factor, in which case κ_4 will be inversely scaled by the same factor, but in the way we develop the theory this arbitrary scaling will be cancelled. The equilibrium value of the order parameter at zero temperature can be obtained from minimization of the potential energy:

$$\frac{\partial E}{\partial \eta} = -\kappa_2 \eta + \kappa_4 \eta^3 = 0 \Rightarrow \eta_0^2 = \kappa_2 / \kappa_4 \quad (12.40)$$

An example of this energy function is shown in Fig. 12.23, where calculations of $E(\eta)$ for quartz are fitted by a function of the form of eqn 12.39.

12.5.3 Phonon free energy

We now proceed by considering the phonon free energy (Chapter 9)

$$F_{\text{ph}} = k_B T \sum_k \ln[2 \sinh(\hbar \omega_k / 2k_B T)] \quad (12.41)$$

For the moment we will make the high-temperature approximation appropriate for $k_B T > \omega_k$, to give

$$\begin{aligned} F_{\text{ph}} &= k_B T \sum_k \ln(\hbar \omega_k / k_B T) \\ &= 3RT \ln(\hbar \langle \tilde{\omega} \rangle / k_B T) \end{aligned} \quad (12.42)$$

We have now replaced the sum over all modes by average values, so that we have from eqn 12.38

$$\langle \tilde{\omega} \rangle^2 = \langle \omega \rangle^2 + \frac{1}{2} \langle \alpha \rangle \eta^2 \quad (12.43)$$

The phonon free energy can be written as a Taylor expansion in the order parameter about the point $\eta = 0$:

$$F_{\text{ph}}(\eta) = F_{\text{ph}}(\eta = 0) + \frac{1}{2} \eta^2 \left(\frac{\partial^2 F_{\text{ph}}}{\partial \eta^2} \right)_{\eta=0} + \dots \quad (12.44)$$

There is no linear term because $\partial F / \partial \eta = 0$ at equilibrium, and we will not consider the higher-order terms. We have the following differentials:

$$\frac{\partial F_{\text{ph}}}{\partial \eta} = \frac{3RT}{\langle \tilde{\omega} \rangle} \frac{\partial \langle \tilde{\omega} \rangle}{\partial \eta} \quad (12.45)$$

$$\frac{\partial^2 F_{\text{ph}}}{\partial \eta^2} = \frac{3RT}{\langle \tilde{\omega} \rangle} \left[\frac{\partial^2 \langle \tilde{\omega} \rangle}{\partial \eta^2} - \frac{1}{\langle \tilde{\omega} \rangle} \left(\frac{\partial \langle \tilde{\omega} \rangle}{\partial \eta} \right)^2 \right] \quad (12.46)$$

From eqn 12.43 we can obtain the following differentials of the frequencies:

$$\frac{\partial \langle \tilde{\omega} \rangle}{\partial \eta} = \frac{\langle \omega \rangle \eta}{2 \langle \tilde{\omega} \rangle}; \quad \left(\frac{\partial \langle \tilde{\omega} \rangle}{\partial \eta} \right)_{\eta=0} = 0 \quad (12.47)$$

$$\frac{\partial^2 \langle \tilde{\omega} \rangle}{\partial \eta^2} = \frac{\langle \omega \rangle}{2 \langle \tilde{\omega} \rangle} - \frac{\langle \omega \rangle \eta}{2 \langle \tilde{\omega} \rangle^2} \frac{\partial \langle \tilde{\omega} \rangle}{\partial \eta}; \quad \left(\frac{\partial^2 \langle \tilde{\omega} \rangle}{\partial \eta^2} \right)_{\eta=0} = \frac{\langle \omega \rangle}{2 \langle \tilde{\omega} \rangle} \quad (12.48)$$

As a result we can write

$$\left(\frac{\partial^2 F_{\text{ph}}}{\partial \eta^2} \right)_{\eta=0} = \frac{3\langle \alpha \rangle RT}{2\langle \omega \rangle^2} \quad (12.49)$$

and

$$F_{\text{ph}}(\eta) = F_{\text{ph}}(\eta = 0) + \frac{3\langle \alpha \rangle RT}{4\langle \omega \rangle^2} \eta^2 \quad (12.50)$$

12.5.4 Full free energy and the Landau free energy function

We now combine the phonon free energy with the potential energy to obtain

$$\begin{aligned} F(\eta) &= F_{\text{ph}}(\eta) + E(\eta) \\ &= F_{\text{ph}}(\eta = 0) + \frac{3\langle \alpha \rangle RT}{4\langle \omega \rangle^2} \eta^2 - \frac{1}{2} \kappa_2 \eta^2 + \frac{1}{4} \kappa_4 \eta^4 \\ &= F_{\text{ph}}(\eta = 0) + \frac{1}{2} \left(\frac{3\langle \alpha \rangle RT}{2\langle \omega \rangle^2} - \kappa_2 \right) \eta^2 + \frac{1}{4} \kappa_4 \eta^4 \\ &= F_{\text{ph}}(\eta = 0) + \frac{1}{2} \left(\frac{3\langle \alpha \rangle R}{2\langle \omega \rangle^2} \right) \left(T - \frac{2\kappa_2 \langle \omega \rangle^2}{3\langle \alpha \rangle R} \right) \eta^2 + \frac{1}{4} \kappa_4 \eta^4 \end{aligned} \quad (12.51)$$

This looks exactly like the Landau free energy function (eqn 12.8)

$$F(\eta) = F(\eta = 0) + \frac{1}{2} a (T - T_c) \eta^2 + \frac{1}{4} \eta^4 \quad (12.52)$$

with

$$T_c = \frac{2\kappa_2 \langle \omega \rangle^2}{3\langle \alpha \rangle R} \quad (12.53)$$

Since κ_2 is the negative of the square of the harmonic frequency of the soft mode, this equation for T_c is exactly the same as eqn 12.34 obtained earlier from the model of the soft mode. We have, however, achieved more than simply replicating the earlier result. We have shown how Landau theory falls out of the lattice dynamics, which gives some justification for the use of Landau theory in the study of displacive phase transitions. We have also obtained microscopic interpretations of the coefficients in the Landau free-energy function.

12.5.5 Low-temperature behaviour

The theory above was developed assuming that we could make the high-temperature approximation. Clearly this is not appropriate at low temperatures. In principle we can take the complete free energy function and expand this in terms of η^2 following the same procedure. We would end up with something similar, albeit with more complicated differentials, and the final equation would not end up looking exactly like Landau theory. In this sense we can deduce that the standard Landau theory is itself a high-temperature approximation, and we already know that at low temperatures it is not appropriate.

Rather than do a complete analysis of the low-temperature behaviour, we will restrict ourselves to $T = 0$ K. Here the free energy is the potential energy plus the zero point phonon motion, giving

$$F(\eta, T = 0) = \frac{3}{2} N_A \hbar \langle \tilde{\omega} \rangle - \frac{1}{2} \kappa_2 \eta^2 + \frac{1}{4} \kappa_4 \eta^4 \quad (12.54)$$

To obtain the equilibrium value of η at $T = 0$ K we calculate the differential of F , using the results for the differential of $\langle \tilde{\omega} \rangle$ given above:

$$\begin{aligned} \left. \frac{\partial F}{\partial \eta} \right|_{T=0} &= \frac{3}{2} N_A \hbar \frac{\partial \langle \tilde{\omega} \rangle}{\partial \eta} - \kappa_2 \eta + \kappa_4 \eta^3 \\ &= \left(3 N_A \hbar \frac{\langle \alpha \rangle}{4 \langle \tilde{\omega} \rangle} - \kappa_2 \right) \eta + \kappa_4 \eta^3 \end{aligned} \quad (12.55)$$

Setting the differential to zero allows us to obtain the equilibrium zero-temperature value of the order parameter, η_0 :

$$\begin{aligned} \left(3 N_A \hbar \frac{\langle \alpha \rangle}{4 \langle \tilde{\omega} \rangle} - \kappa_2 \right) \eta + \kappa_4 \eta^3 &= 0 \\ \Rightarrow \eta_0^2 &= \frac{\kappa_2}{\kappa_4} - \frac{3 N_A \hbar \bar{\alpha}}{4 \kappa_4 \langle \tilde{\omega} \rangle} \end{aligned} \quad (12.56)$$

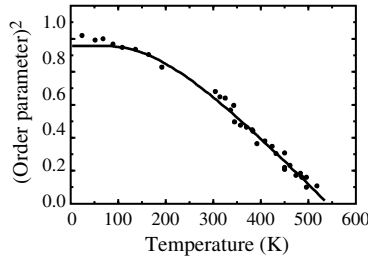


Fig. 12.24 Temperature dependence of the square of the order parameter associated with the displacive phase transition in the aluminosilicate $\text{CaAlSi}_3\text{O}_8$. The curve represents a fit of the theory of this chapter to the data, taking account of the quantum terms at low temperature.

Note that because we are not now expanding around $\eta = 0$, the equation for η_0 contains $\langle \tilde{\omega} \rangle$ rather than $\langle \omega \rangle$; the former is the frequency modified by the non-zero value of η , and the latter is the frequency extrapolated from the high-temperature phase. This result shows that the effect of zero point motion is to lower the zero-temperature value of η below that given by minimization of $E(\eta)$ alone.

An example of the complete quantum mechanical version of this theory fitted to experimental data for the order parameter is shown in Fig. 12.24. This clearly shows a linear dependence of η^2 close to T_c , and the expected curvature away from this linear relation at low temperatures.

Summary of chapter

- **Displacive phase transitions** occur on changing temperature or pressure, and involve small symmetry-breaking displacements of atoms that can be reversed on changing temperature or pressure.
- The symmetry-breaking atomic displacements associated with **ferroelectric** phase transitions lose the centre of symmetry and give rise to the formation of an electric dipole moment in the unit cell. The resultant dielectric polarization of the crystal can be reversed by application of an electric field. The most common examples of ferroelectric phase transitions are found in the perovskite family of structures, e.g. PbTiO_3 and BaTiO_3 . Ferroelectric phase transitions are accompanied by a divergence of the dielectric constant.
- The atomic displacements associated with some displacive phase transitions are different in neighbouring unit cells, leading to a new lattice

whose unit cells have volumes that are multiples of the initial unit cell volumes. One example is the displacive phase transition in SrTiO_3 , which involves rotations of the TiO_6 octahedra.

- **Ferroelastic** phase transitions are those for which the primary instability is a shear strain of the lattice (that is, the change in symmetry of the crystal must be equivalent to the change in symmetry associated with the shear strain). Ferroelastic phase transitions are accompanied by the softening of an elastic constant.
- The symmetry-breaking distortions associated with a displacive phase transition can be described by an overall **order parameter** (or a small number of independent order parameters). Examples of order parameters are the size of the dielectric polarization associated with a ferroelectric phase transition (in this case it is possible to define separate order parameters for the three possible vector components of the polarization), or the angle of polyhedral rotation.
- Phase transitions in which the order parameter falls continuously to zero at the transition temperature are defined as **second-order phase transitions**. Those for which there is a discontinuous change in the order parameter at the transition temperature are defined as **first-order phase transitions**. First-order phase transitions also have a discontinuous change in the entropy. For many second-order displacive phase transition the square of the order parameter falls linearly to zero at the transition temperature.
- The thermodynamics associated with the phase transition can be described by an expansion of the free energy in what is known as **Landau theory**. This can describe both first and second-order phase transitions, and is able to incorporate variables such as strain. Although Landau theory is a phenomenological model, it properly describes symmetry in the way that it incorporates different variables, and as a result has some degree of predictive power. Landau theory's main failing is to describe the behaviour of the system at low temperatures.
- An important component of the mechanism of a displacive phase transition is the **soft mode**, namely a phonon mode whose frequency falls to zero at the phase transition. ω^2 represents the restoring force against a set of atomic displacements, which vanishes to allow the phase transition to take place. The temperature dependence of the soft mode arises from **anharmonic phonon interactions**.
- The thermodynamics of the phase transition can be represented by incorporating the dependence of the phonon frequencies on the order parameter, giving results for the transition temperature that are in accord with the transition temperature calculated by anharmonic phonon theory. The Landau free energy function can be obtained from an expansion of the phonon free energy and lattice energy as a power series in the order parameter in the high-temperature limit.
- The low-temperature behaviour of the order parameter is determined by the quantum mechanics of phonons. At absolute zero, the dependence of the zero point motions on the order parameter will cause a reduction of the order parameter, and in extreme cases can suppress the phase transition and stabilize the high-temperature phase.

Further reading

The topic of phase transitions is not usually covered in textbooks on solid state physics other than in discussions of dielectric properties and magnetism; both Kittel (1996) and Ashcroft and Mermin (1976) give introductions to the subject. Dove (1993, 1997), Rao and Rao (1978), and Salje (1990) give complementary comprehensive treatments at an introductory level. A more advanced treatment is given by Bruce and Cowley (1981). The statistical mechanics of phase transitions are discussed by Yeomans (1992). Landau theory is discussed to an advanced level by Tolédano and Tolédano (1987). Some wider issues are discussed in a review article written by the author (M. T. Dove, 1997, *American Mineralogist*, **82**, 213–44.)

Exercises

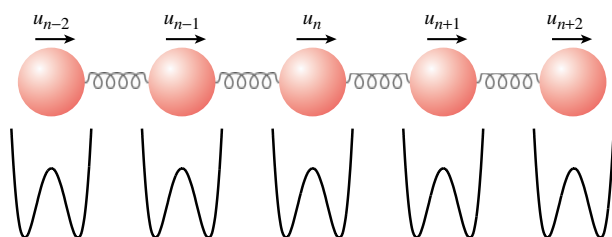
- (12.1) A **tricritical** phase transition is on the border between first-order and second-order behaviour. Such a phase transition might be described by a Landau free energy function of the form of eqn 12.19 with $b = 0$. Obtain expressions for the temperature dependence of the order parameter, susceptibility, and heat capacity for $T < T_c$, and contrast the results with those for second-order and first-order phase transitions (Sections 12.3.3 and 12.3.4 respectively).
- (12.2) Consider a first-order phase transition described by Landau theory, eqn 12.19. At the transition temperature, there are minima of $F(\eta)$ at a non-zero value of η^2 with $F(\eta) = 0$. By combining this condition with the condition that $\partial F/\partial \eta = 0$ at the minima, show that the minimum occurs for $\eta^2 = 3b/4c$. Use this result to show that the transition temperature is given by eqn 12.21. Show that eqn 12.20 is the stable solution of $\partial F/\partial \eta = 0$ for $T < T_c$.
- (12.3) Show that at the temperature $T_0 + b^2/4ac$ the Landau free energy function for a first-order phase transition, eqn 12.19, develops a new minimum at $\eta^2 = b^2/4c$.
- (12.4) Obtain the parameters T_c , b/a , and c/a from the data for the temperature dependence of the ferroelectric polarization in PbTiO_3 given in Fig. 2.30 ($\eta = P$). This will require the use of the minimization function in a spreadsheet program together with judicious use of the data. Use the results to plot the form of $F(P)$ for various temperatures.
- (12.5) For some phase transitions, symmetry allows the Landau free energy to be written as
- $$F(\eta) = \frac{1}{2}a(T - T_c)\eta^2 + \frac{1}{4}\eta^4 + \frac{1}{2}\lambda\epsilon_s\eta + \frac{1}{2}C_{el}\epsilon_s^2$$
- where ϵ_s is the spontaneous strain. Show (a) that the equilibrium condition $\partial F/\partial \epsilon_s = 0$ gives $\epsilon_s = -\lambda\eta/2C_{el}$; (b) that this coupling leads to a new transition temperature at $T_c + \lambda^2/4aC_{el}$; (c) that the corresponding elastic constant falls to zero at the phase transition temperature.
- (12.6) It is often found that a doubly-degenerate phonon will split at a phase transition to give two new frequencies as $\omega^2 = \omega_0^2 \pm \gamma\eta$. Show that
- $$\left. \frac{\partial \omega}{\partial \eta} \right|_{\eta=0} = \frac{\gamma}{2\omega_0}$$
- $$\left. \frac{\partial^2 \omega}{\partial \eta^2} \right|_{\eta=0} = \pm \frac{\gamma^2}{4\omega_0^3}$$
- Substitute these contributions into eqns 12.44–12.46 to show that each split mode gives a contribution to the free energy in the high-temperature limit of
- $$f = -\frac{\gamma^2 k_B T}{4\omega_0^3} \eta^2$$
- Demonstrate that this has the effect of raising the transition temperature.
- (12.7) Equation 12.53 gives T_c obtained in the high-temperature approximation, namely for $k_B T > \hbar\langle\omega\rangle$. By comparing eqns 12.53 and 12.56, show that if the calculated value of $k_B T_c < \hbar\langle\omega\rangle/2$ the phase transition is suppressed through quantum fluctuations.

- (12.8) Show that at zero temperature, eqn 12.32 for the soft mode frequency is given by

$$\omega^2 = \omega_0^2 + \frac{3N_A \bar{\alpha}}{4\langle\omega\rangle}$$

Equation 12.34 gives an alternative formulation of T_c in terms of the squares of the harmonic frequency of the soft mode at zero temperature, ω_0^2 , again calculated within the high-temperature limit. Show that if the calculated value of $k_B T_c < \hbar\langle\omega\rangle/2$, the soft mode frequency is prevented from becoming unstable at zero temperature.

- (12.9) The figure below shows a simple model for a phase transition.



Each atom interacts with its neighbours through harmonic forces, characterized by the force constant J , but also is subject to forces through a local potential energy function of the form

$$E(u) = -\frac{\kappa_2}{2}u^2 + \frac{\kappa_4}{4}u^4$$

Show that the energy of atom j in a one-dimensional chain can be written as

$$E_j = \frac{2J - \kappa_2}{2}u_j^2 - Ju_j(u_{j+1} + u_{j-1}) + \frac{\kappa_4}{4}u_j^4$$

Obtain the dispersion curve for the harmonic component of this energy.

The anharmonic component can be approximated as $u^4 \simeq 3u^2\langle u^2 \rangle$, and we can approximate $\langle u^2 \rangle = k_B T / \omega_0^2$. Modify the dispersion curve to incorporate this quasi-harmonic term, and show that the model gives a phase transition at $k_B T_c = \kappa_2 \omega_0^2 / 3\kappa_4$.

- (12.10) Insights into the rotational phase transition in the perovskite structure can be gained by treating the octahedra as nearly rigid objects. If each octahedron is assigned a single variable to describe its orientation, θ , two neighbouring octahedra have lowest energy if they rotate by exactly the same amount in opposite directions. Show that the harmonic energy between two neighbouring octahedra can be written as

$$E(i, j) = \frac{1}{2} K (\theta_i + \theta_k)^2$$

For a phase transition to occur, it can be presumed that each octahedron also experiences a local double-well potential of the form

$$E(\theta) = -\frac{\kappa_2}{2}\theta^2 + \frac{\kappa_4}{4}\theta^4$$

Show that when the harmonic term is combined with the interactions between neighbours, a soft mode is obtained at $\mathbf{k} = (\frac{1}{2}, \frac{1}{2}, 0)$.

- (12.11) The Landau theory for the ferroelectric phase transition in the perovskite structure can be written in the following form to account for the components of the polarization along the three axes:

$$F = \frac{1}{2}a(T - T_c)(P_x^2 + P_y^2 + P_z^2) + \frac{1}{4}b(P_x^4 + P_y^4 + P_z^4) + \frac{1}{4}\gamma(P_x^2 + P_y^2 + P_z^2)^2$$

Describe the conditions under which the stable solutions are

- $P_x \neq 0$ and $P_y = P_z = 0$
- $P_x = P_y \neq 0$ and $P_z = 0$
- $P_x = P_y = P_z \neq 0$.

- (12.12) A given material undergoes a displacive phase transition at $T_c = 950$ K. The vibrations have an average frequency of 10 THz, and the phase transition causes the vibrational frequencies to increase by an average of 0.02 THz on cooling to low temperature. The phase transition is accompanied by a step change in the heat capacity of $0.124 \text{ J mol}^{-1} \text{ K}^{-1}$. Calculate (a) the coefficients of the Landau free energy for the phase transition on the assumption that the data can be interpreted in the high-temperature limit; (b) the change in the value of the order parameter at low temperature due to the effects of zero point motions. (Note: you will need to determine values of κ_2 and κ_4 ; their exact values will depend on how you chose to normalize the value of the order parameter at low temperature.)

Use a spreadsheet to minimize the free energy with respect to the order parameter for a range of temperatures in order to obtain the temperature dependence of the order parameter, using the values of the parameters obtained in the first part of the question. From your plot, ascertain the validity of the high-temperature approximation for the analysis of the free energy around T_c . At what temperatures does a plot of the square of the order parameter against temperature deviate from a nearly linear relationship. Is this temperature in accord with the average vibrational frequency?